Name: _____

Investigating radiation

core practical

Date:

Time:

Total marks available: 56

Total marks achieved: _____

Questions

Q1.

Figure 12 shows two cans, a radiant heater and some other apparatus.

The cans absorb thermal radiation from the heater.

One can has a matte black surface and the other can has a shiny silver surface.

Both cans contain water at the same temperature.

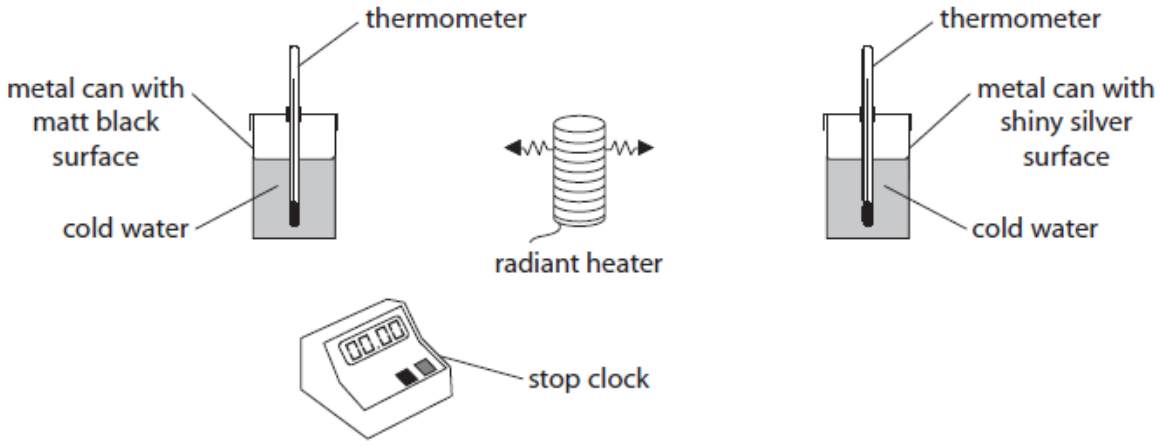

Figure 12

Describe how a student could use the apparatus in Figure 12 to determine which can is the better absorber of thermal radiation.

(4)

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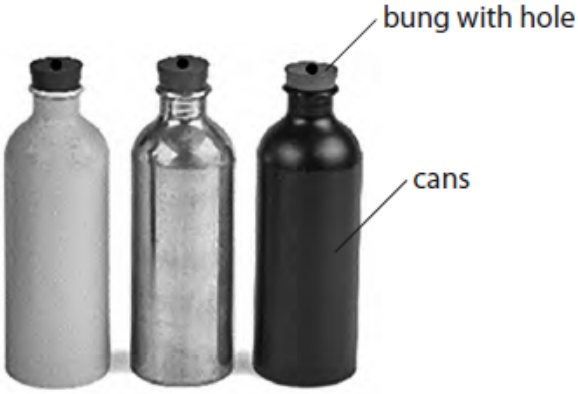
(Total for question = 4 marks)

Q2.

Figure 7 shows some apparatus.

small infrared heating lamp



set of three cans of the same size and material but of different surfaces

Figure 7

Describe an investigation to find out how the nature of a surface affects the amount of thermal energy absorbed by the surface.

You should use the apparatus in Figure 7 and any additional items you choose.
Each can in Figure 7 has a bung in the top with a hole in it.

You may use a diagram if it helps your answer.

(6)

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(Total for question = 6 marks)

Q3.

A student investigates how the surface of an object affects the radiation it emits.

Figure 7 shows the equipment he uses.

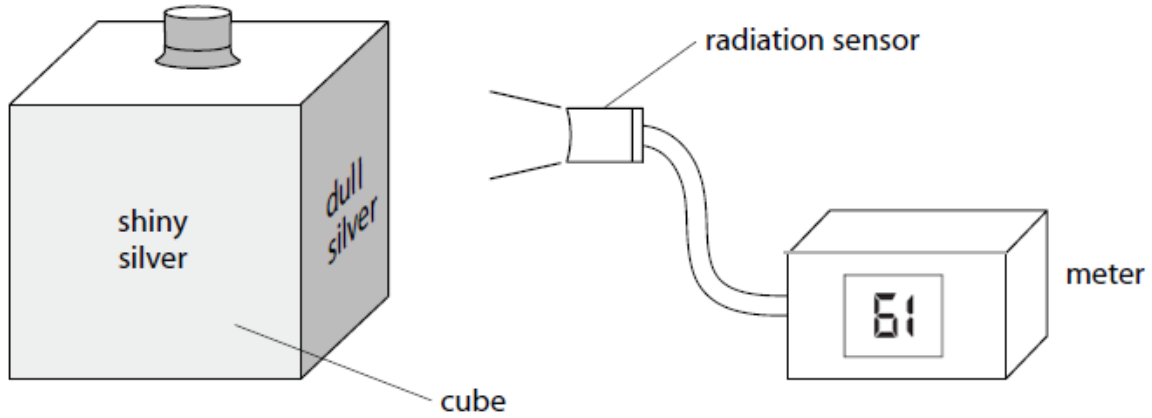


Figure 7

The cube has four different surfaces.

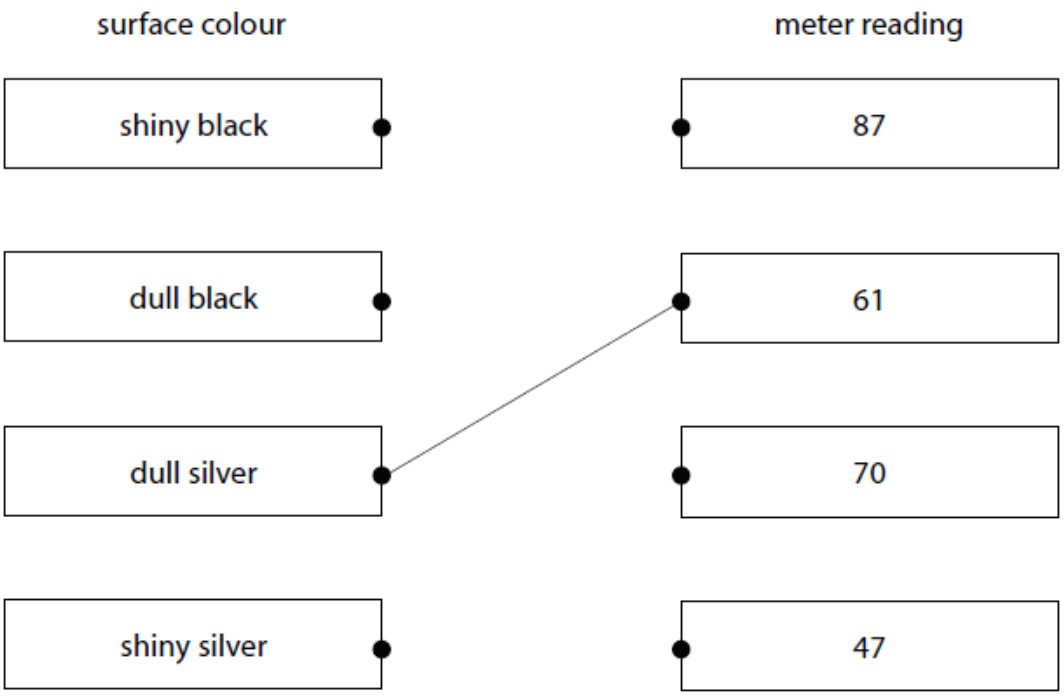
He fills the cube with boiling water so that the temperature of each surface is the same.

He uses the radiation sensor to measure the radiation emitted from each surface.

(i) His readings are shown.

Draw a line from each surface colour to its correct meter reading. One has been done for you.

(2)



(ii) The temperature of each surface is the same.

Give a reason why the radiation sensor gives a different reading for each surface.

(1)

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(Total for question = 3 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 6 shows two solid metal balls, **P** and **Q**.

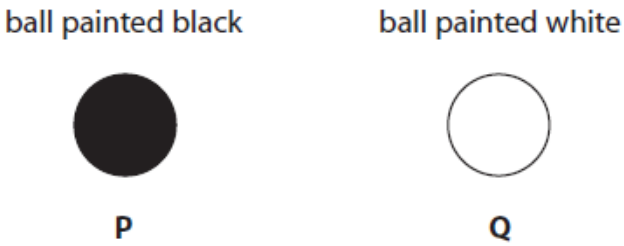

Figure 6

P and **Q** are made from the same metal and have the same radius.

P is painted black and **Q** is painted white.

Each ball is heated to a different temperature.

The balls then cool in the same room.

The graph in Figure 7 shows how the temperature of each ball changes with time.

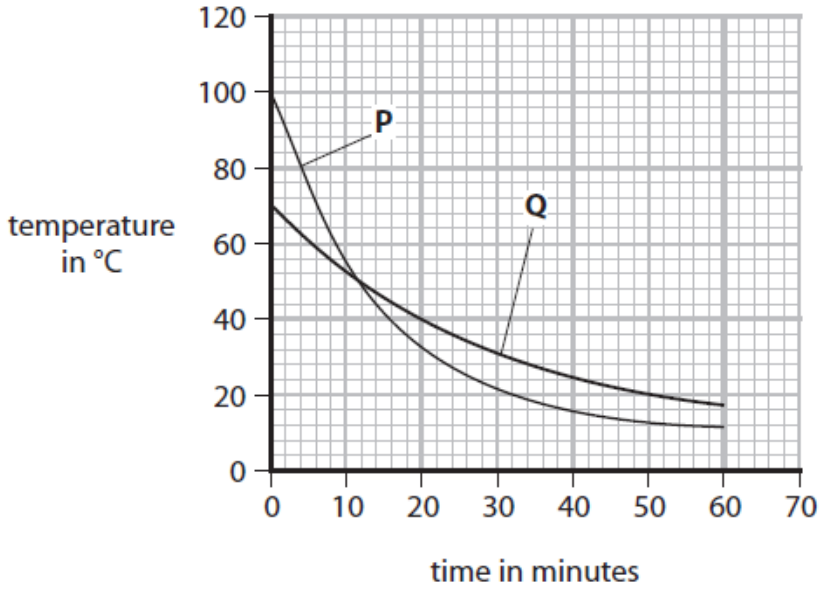

Figure 7

(i) Use the graph in Figure 7 to determine the time when **P** and **Q** were at the same temperature.

Show your working on the graph.

(2)

time = ... minutes

(ii) Which of these temperatures is most likely to be room temperature, as shown by the graph in Figure 7?

(1)

☒ **A** 100 °C

☒ **B** 70 °C

☒ **C** 10 °C

☒ **D** 0 °C

(iii) Explain why the curve for **P** is different from the curve of **Q**. Use information from Figure 6 and Figure 7 to
help your answer.

(2)

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(Total for question = 5 marks)

Q5.

Some students repeat Herschel's experiment.

(a) They place four identical thermometers, **P**, **Q**, **R**and **S**, in the shade.

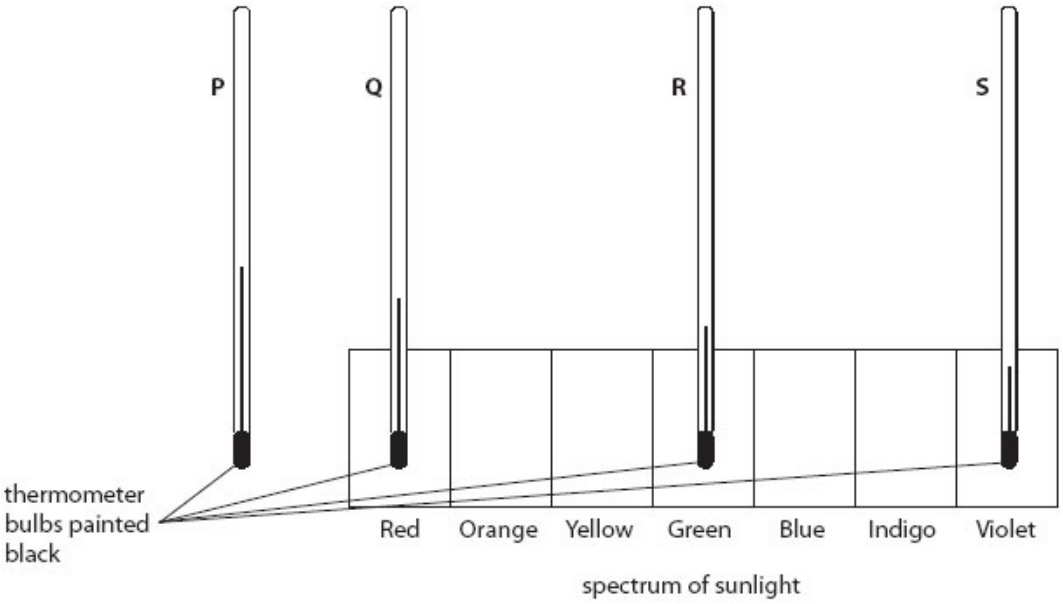
The table shows some of the readings on the thermometers in the shade.

Complete the table to show the reading on thermometer **Q**.

(1)

	thermometers in the shade				
	P	**Q**	**R**	**S**	
temperature / °C	18		18	18	

(b) Then the four thermometers are placed in the spectrum of sunlight as shown.



(i) The bulbs of the thermometers are painted black.

State why the bulbs of the thermometers are painted black.

(1)

.....

.....

(ii) The table shows the readings on the thermometers in the spectrum of sunlight.

	thermometers in the spectrum of sunlight			
	P	**Q**	**R**	**S**
temperature / °C	25	23	21	20

Describe conclusions that can be made from the students' results.

(2)

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(iii) State why the students put the four thermometers together in the shade before placing them in the spectrum of sunlight.

(1)

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(c) Another scientist, Ritter, discovered ultraviolet radiation.

(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

Too much exposure to ultraviolet radiation may cause

(1)

☒ **A** deafness

☒ **B** heating of internal body cells

☒ **C** damage to the eyes

☒ **D** damage to the bone cells

(ii) Three signals, ultraviolet, visible light and infrared, are sent from the surface of the Moon to an orbiting spacecraft.

The three signals are sent at the same time.

Which of these is correct for the signals arriving at the spacecraft?

Put a cross (☒) in the box next to your answer.

(1)

☒ **A** the visible light signal arrives first

☒ **B** the ultraviolet signal arrives first

☒ **C** the infrared signal arrives first

☒ **D** all three signals arrive at the same time

(iii) Describe **one**use of ultraviolet radiation.

(2)

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(Total for Question is 9 marks)

Q6.

A student wants to use sunlight to heat water in a container.

(i) The student can choose from four containers.

The containers:

- are all the same size
- have different colours
- have different surfaces.

Which container is the best absorber of thermal energy?
Put a cross (☒) in the box next to your answer.

(1)

	container colour	container surface
☐ **A**	light	shiny
☐ **B**	light	rough
☐ **C**	dark	shiny
☐ **D**	dark	rough

(ii) The student places the container in bright sunlight. After half an hour, the temperature of the water is 50°C. The water then stays at 50°C, even though the container is still in bright sunlight.

Which of these explains why the temperature stays at 50°C?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** The water in the container is no longer absorbing thermal energy.

☐ **B** The container is losing thermal energy faster than it is absorbing it.

☐ **C** The container is losing thermal energy at the same rate it is absorbing it.

☐ **D** The water in the container has started to boil.

Q7.

Equal volumes of hot water are added to two cans.
The cans are identical apart from their surfaces.
One can has a black surface and the other can has a silver surface.

The cans are left to cool and their temperatures are monitored.

The graph in Figure 6 shows the results.

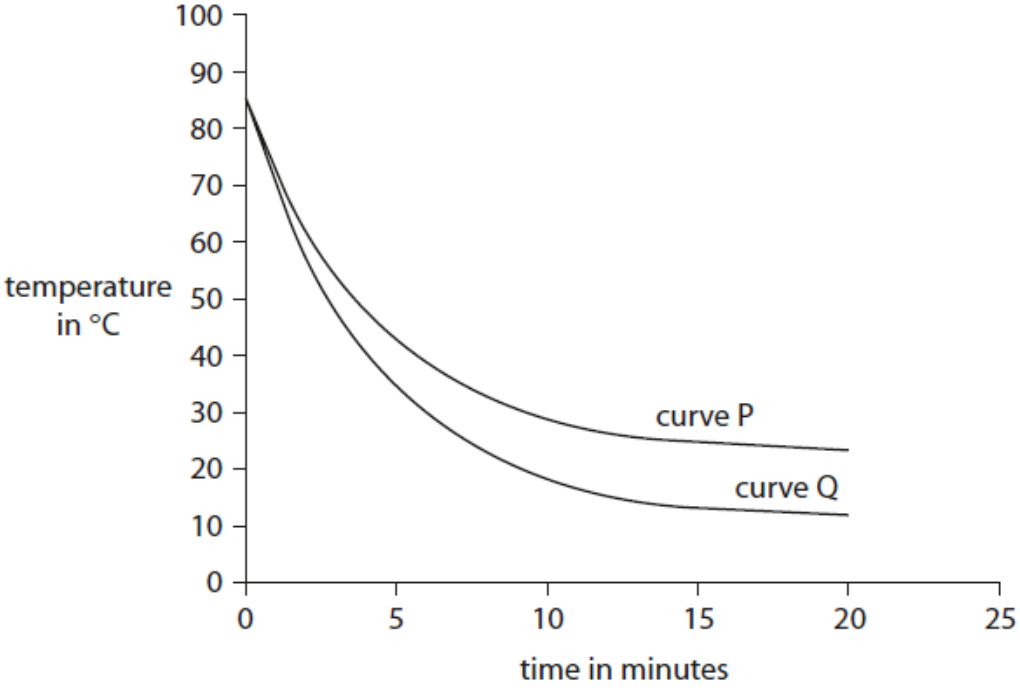

Figure 6

Explain, using evidence from the graph, which curve is for the black can and which curve is for the silver can.

(2)

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(Total for question = 2 marks)

Q8.

Figure 16 shows two iron spheres, **P** and **Q**, near to a radiant heater.

P is painted black and **Q** is painted white.

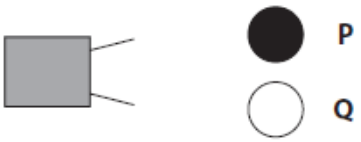

Figure 16

Each sphere is the same distance away from the heater.

The spheres have the same radius.

The heater is switched on and the spheres heat up.

The temperature of each sphere is monitored.

(i) Explain why the temperature of sphere **P** increases at a faster rate than the temperature of sphere **Q**.

(2)

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(ii) The heater remains switched on.

Figure 17 shows how the temperature of sphere **P** changes with time.

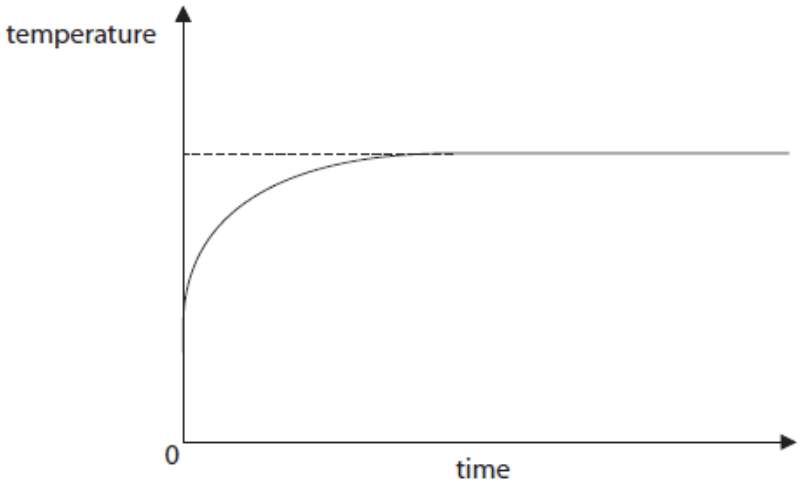

Figure 17

Explain why the temperature of **P** reaches a constant value, even though the heater remains switched on.

(4)

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(Total for question = 6 marks)

Q9.

A student investigates how different surfaces radiate energy as they cool.

Figure 9 shows some of the apparatus used in a part of the investigation.

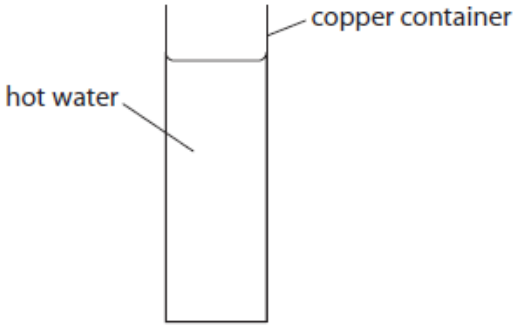

Figure 9

Describe how the student could collect data to show how the rate of cooling of the container and water change with time.

(2)

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(Total for question = 2 marks)

Q10.

A copper can, painted black, contains boiling water at 100 °C.

The can is left to cool and a measurement of the water temperature is taken every 5 minutes.

Figure 3 shows the measurements.

time in minutes	temperature in °C
0	100
5	74
10	60
15	56
20	37
25	30
30	25

Figure 3

(i) Two points, shaded in the table, have not been plotted.

Plot these two points on the graph, in Figure 4.

(2)

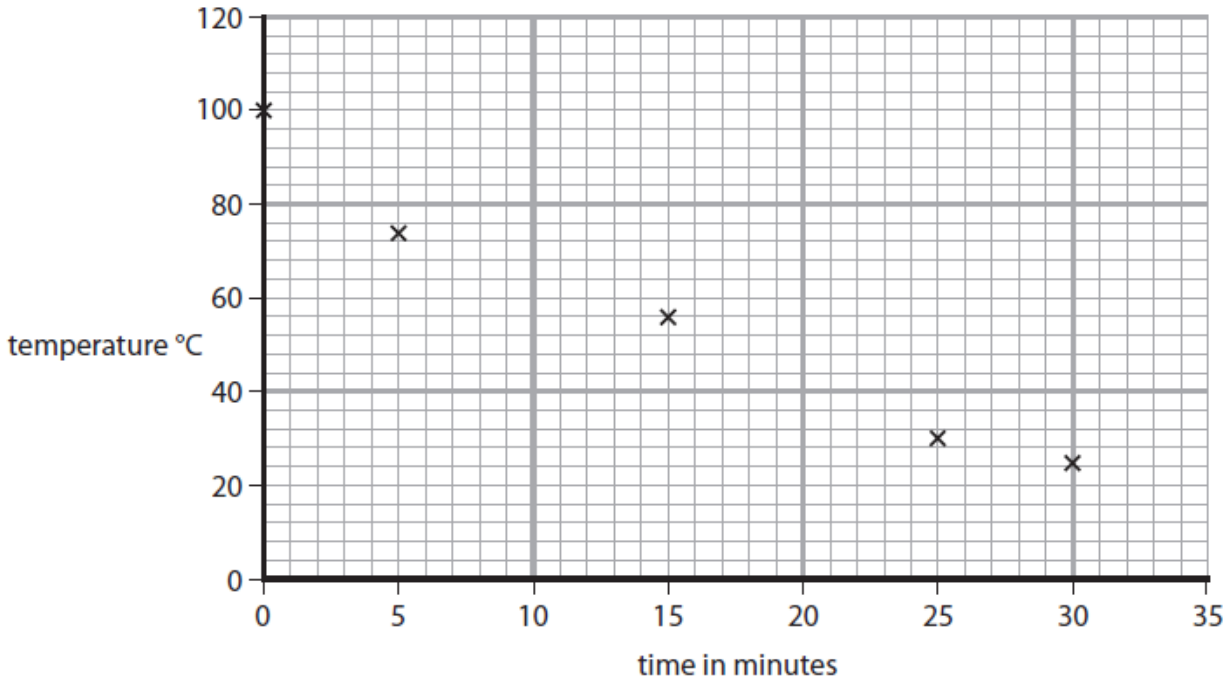

Figure 4

(ii) One of the points on the graph in Figure 4 is anomalous.

Circle the anomalous point.

(1)

(iii) Draw the best fit curve on the graph in Figure 4.

(1)

(iv) An identical can contains the same amount of boiling water.

This can has a shiny silver surface.

The measurements are repeated with this can and a new curve is drawn.

State how the cooling curve would be different from the curve in the graph in Figure 4.

(1)

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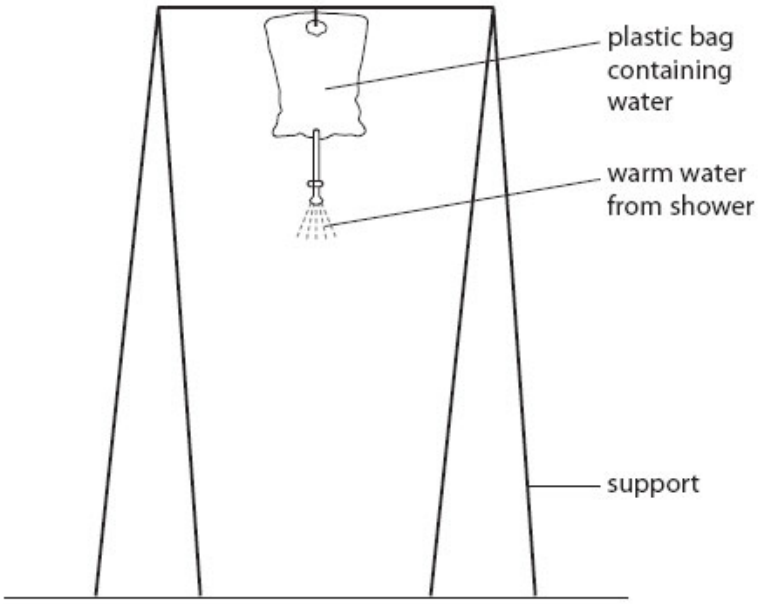
(Total for question = 5 marks)

Q11.

The following arrangement is used as a solar powered shower.



The bag is left out in the sunlight during the day.

(i) Explain what colour the bag should be to heat the water to the highest temperature.

(2)

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(ii) On a sunny day the bag is filled with cold water.
Explain why the temperature of the water increases and then stays constant.

(3)

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Q12.

Figure 9 shows a metal cube filled with hot water.

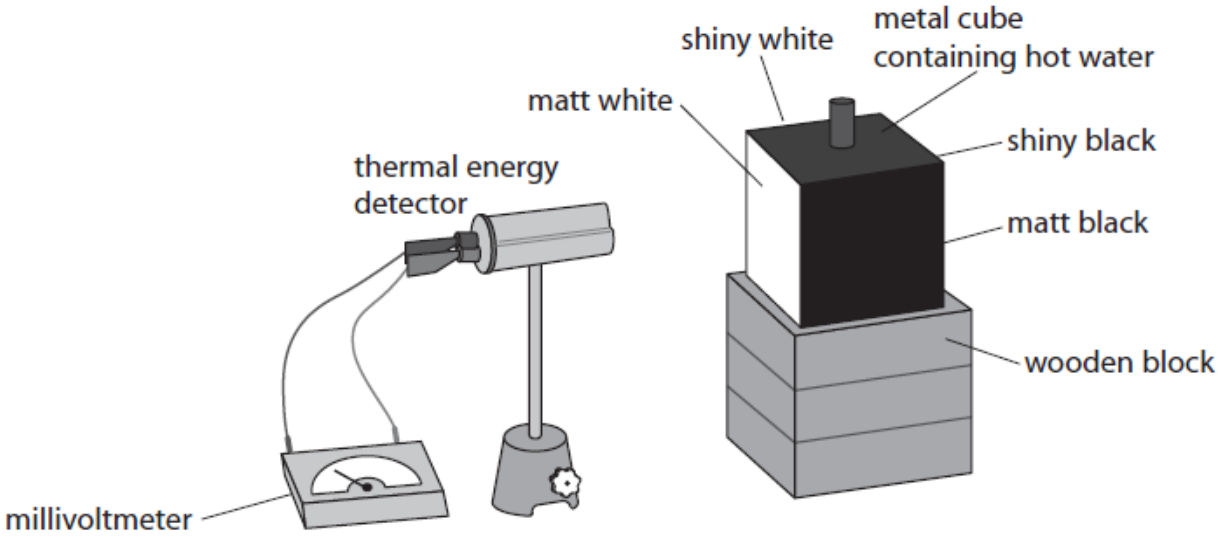

Figure 9

Four sides of the cube have different surfaces, as shown in Figure 9.

The four surfaces are at the same temperature.

The thermal energy radiated by each side of the box is measured using a thermal energy detector connected to a millivoltmeter.

The detector is moved to get a reading of the thermal energy emitted from each side of the box.

Figure 10 shows the table of results.

type of surface	millivoltmeter reading in mV
matt white	32
shiny white	20
shiny black	
matt black	55

Figure 10

(i) Suggest a possible millivoltmeter reading for the shiny black surface.

(1)

..... mV

(ii) State what must be kept the same to take the measurement for each surface.

(1)

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(iii) Suggest why the cube is placed on a block of wood.

(1)

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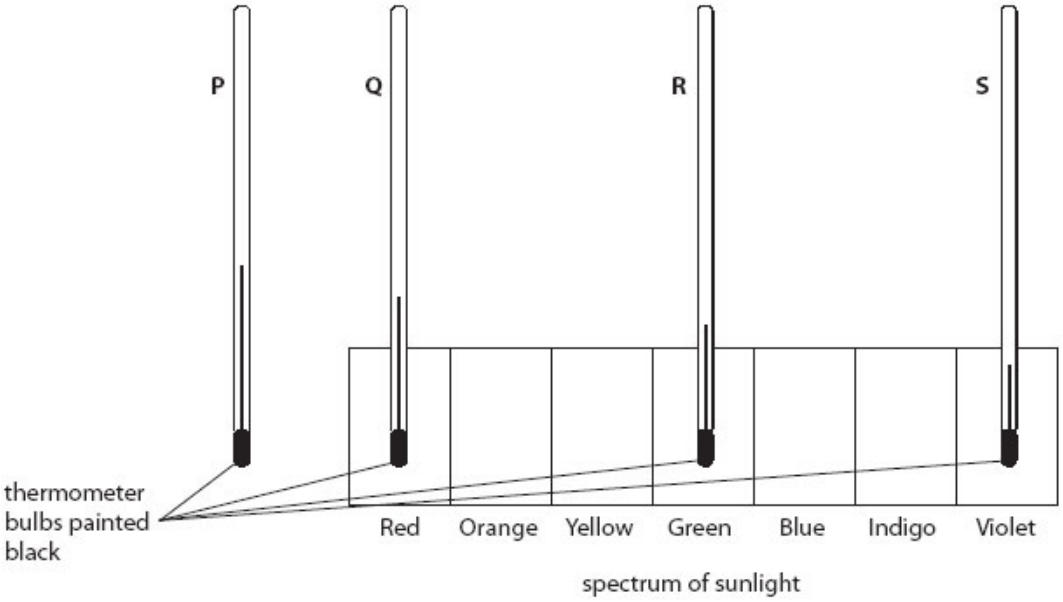
(Total for question = 3 marks)

Q13.

Some students repeat Herschel's experiment.

Then the four thermometers are placed in the spectrum of sunlight as shown.



(i) The bulbs of the thermometers are painted black.

State why the bulbs of the thermometers are painted black.

(1)

.....

.....

(ii) The table shows the readings on the thermometers in the spectrum of sunlight.

	thermometers in the spectrum of sunlight			
	P	**Q**	**R**	**S**
temperature / °C	25	23	21	20

Describe conclusions that can be made from the students' results.

(2)

.....

.....

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.....

(iii) State why the students put the four thermometers together in the shade before placing them in the spectrum
of sunlight.

(1)

.....

.....

Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
	a description including any four from **one** method **Using temperature change method:** any one control (1) start the stopclock and switch on the heater (at the same time) (1) measure over a certain time (1) measure final temperature of each (1) largest temperature difference/rise gives better absorbing surface (1)	allow testing each can separately (two experiments) same distance from heater same volume/mass of water allow same volume/surface area of cans lids on both cans start the stopclock and measure initial temperature (at the same time) Accept 'highest temperature is better'	**(4)** **AO1**

	Using time change method: any one control (1)	same distance from heater same volume/mass of water allow same volume/surface area of cans lids on both cans	
	switch on the heater and start the stopclock (at the same time) (1)		
	measure time taken (1)		
	to reach a certain temperature (1)		
	quickest time to reach a certain temperature gives better absorbing surface (1)	accept 'fastest (to reach temperature) is better'	

Q2.

Question Number	Answer	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO targeting: 6 marks AO1 strand 2 (6 marks)** • uses a thermometer • inserted into the can • uses a stopwatch / timer • uses a measuring cylinder / beaker with markings • sets up apparatus with lamp a fixed distance from each can • that distance is the same for each can • the measured volume is the same for each can • a starting temperature is measured • (aiming) to get the same start temperature • each, in turn, is heated for the same time • as measured on a stopwatch • a final (maximum) temperature is read • OR takes a series of temperature readings with time • and plots a graph of temperature(s) against time • the one heating up the most is the best absorber N.B. fully credit any of these in a diagram e.g. bullet points 1-8 may all be seen in a diagram	**(6)**

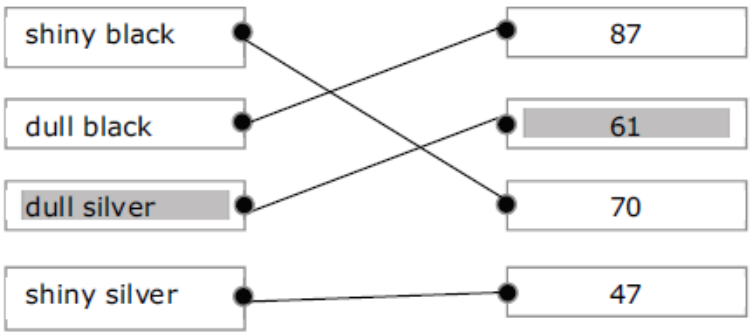
Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1) • Presents a description which is not logically ordered and with significant gaps. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1) • Presents a description of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1) • Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

Summary for guidance

Level	Mark	Additional Guidance	General additional guidance – the decision within levels Eg - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated statement about apparatus or procedure or the science involved	Possible candidate responses (diag?) uses/puts a thermometer in the can/water OR measures a temperature OR uses heater to heat the cans Or puts water in the cans
Level 2	3–4	Additional guidance some procedure detail but with omissions	Possible candidate responses (diag?) uses a thermometer to measure the temperature rise in the cans OR uses the heater to heat the cans for a fixed time Or uses the same amount of water in each can

Level 3	5–6	Additional guidance more detailed and structured procedure	Possible candidate responses (diag?) uses a thermometer to measure the temperature rise in the cans AND uses the heater to heat the cans for a fixed time Or uses the same amount of water in each can

Q3.

Question number	Answer	Mark
(i)	All three correct (2) One or two correct (1)  	(2)

Question number	Answer	Additional guidance	Mark
(ii)	Different surfaces emit (thermal) radiation at different rates	allow reference to surfaces in question	(1)

Q4.

Question number	Answer	Additional guidance	Mark
(i)	line shown on graph from intersection of two curves (1) answer in range 11 – 13 (minutes) inclusive (1)	 award full marks for the correct answer without working	(2)

Question number	Answer	Mark
(ii)	☒ **C** 10 °C	(1)

Question number	Answer	Additional guidance	Mark
(iii)	an explanation linking: the gradient for **P** is greater/steeper than the gradient for **Q** (1) (because) **P**/black is a better emitter (of radiation) (than **Q**/white) (1)	 **P** cools quicker than **Q** **P** loses thermal energy/heat quicker than **Q** allow reverse arguments credit answers in terms of absorption in this context	(2)

Q5.

	Answer	Acceptable answers	Mark
(a)	18 (°C) (1) Ignore any unit given by candidate.		(1)
(b)(i)	(black is the) best absorber (of radiation/heat)(compared to other colours) (1)	(Black/it) absorbs more (Black/it) is a good absorber (of radiation/heat) (black is) good at taking in (radiation/heat) Ignore (black is a) good emitter Ignore light. Reject black attracts heat/radiation	(1)

(b)(ii)	• Heating effect/temperature greatest at/beyond red (1) • (There is) radiation beyond red (end of spectrum)(1)	(idea of) different colours have different heating effects (Radiation from) sunlight causes a heating effect accept reverse argument Infrared/IR (beyond red end of spectrum)	**(2)**
(b)(iii)	Any **one**from • To check the thermometers produce the same readings/ temperature (under the same conditions) (1) • To show that temperature changes.	(To check) they were all at the same temperature (before starting the experiment.) To be able to make a comparison (between shade and light) (To allow them to carry out a) fair test	**(1)**
(c)(i)	C damage to the eyes (1)		**(3)**
(c)(ii)	D all three signals arrive at the same time (1)		**(1)**
(c)iii	Description linking **one**of the following pairs: • security marking (1) • ink absorbs UV and re-radiates (visible) light (1) • fluorescent lamps (1) • coating absorbs UV and reradiates (visible) light (1) • genuine bank notes (1) • watermark absorbs UV and reradiates (visible) light (1) • disinfecting water (1) • UV kills bacteria (1) • sun beds (1) • UV absorbed by (melanin in) skin (1) Any suitable use gains 1 mark Any suitable use + detail gains 2 marks	invisible ink/smart water glows under UV (outside of) lamp glows when hit by UV forgeries/fake bank notes/passports/fingerprints/ body fluids etc markings glow under UV tanning beds tans the skin /the body e.g. disco lighting (1) makes clothing glow (1)	**(2)**

(Total for question = 9 marks)

Q6.

Question Number	Answer	Acceptable answers	Mark
(i)	**D** dark : rough		**(1)**

Question Number	Answer	Acceptable answers	Mark
(ii)	**C** the container is losing thermal energy at the same rate it is absorbing it	Heat for thermal Same amount in same time for same rate	**(1)**

Q7.

Question Number	**Answer**	**Additional guidance**	**Mark**
	an explanation linking **two** from: identifies curve Q is the black can OR that curve P is for the silver can (1) (because, as) black is a better emitter (then) (curve) Q shows a faster rate of cooling (1) OR (curve) Q is steeper (1)	 IGNORE reference to absorb allow Q is at a lower temperature/stays cooler OR P is at a higher temperature /stays hotter may quote numbers from graph e.g. curve Q cools by 75(°C) in 20 minutes but curve P only shows a 60(°C) in the same time	**(2)**

Q8.

	Answer	Additional guidance	Mark
(i)	explanation linking any 2 of: black / P is good absorber / poor reflector (1) white / Q is good reflector poor absorber (1) different colour of surfaces absorb differently (1)	allow any one statement for 2 marks: • black / P **better** absorber • black / P **worse** reflector • white / Q **worse** absorber • white / Q **better** reflector • black / P absorbs (radiation) **and** white / Q reflects (radiation)	**(2)** **AO1**

	Answer	Additional guidance	Mark
(ii)	explanation linking **four** from: P is absorbing radiation/(thermal) energy (1) P is (also) emitting radiation/(thermal) energy (1) (rate of) emission increases with temperature (1) (temperature constant) when (rate of) emission = (rate of) absorption (1) mention of rate in either mp3 or mp4 (1)	allow heat in this context temperature of P constant when rate of emission of radiation = (rate of) absorption of radiation scores 4 marks (rate of) absorption of radiation is greater than (rate of) emission of radiation (so temperature increases) scores MP1 and 2	**(4)** **AO1**

Q9.

Question Number	Answer	Additional guidance	Mark
	a description to include: use a thermometer (1) measure temperature at regular intervals (1)	 temp. sensor datalogger it must be clear that it is a number of readings – not just 2 eg measure temperature over time	**(2)** AO 1 2

Q10.

Question Number	Answer	Additional guidance	Mark
(i)	points plotted to within ± 1 small square (10, 60) (1) (20, 37) (1)		**(2)** AO 2 1

Question Number	Answer	Additional guidance	Mark
(ii)	point at 15,55 circled (1)		**(1)** AO 2 1

Question Number	Answer	Additional guidance	Mark
(iii)	best fit curve passing through most of the points seen, apart from 15,55 (1)	reject tramlining (multiple lines/curves) ignore slight shakiness in drawing	**(1)** AO 3 2a

Question Number	Answer	Additional guidance	Mark
(iv)	the curve will be above the first one (starting at the same point) / it will take longer to cool down / cools more slowly (1)	less steep ignore all comments about heat / the process; **the question is about the curve**	**(1)** AO 2 1

Q11.

	Answer	Acceptable answers	Mark
(i)	An explanation linking the following points: black (1) (because) (good) absorber (of thermal radiation) (1)	 {absorbs / takes in} heat radiation **ignore**references to: attract good emitter light dark / darker	**(2)**
(ii)	an explanation linking any **three**of the following points: • (bag / water) absorbs {thermal energy / heat / radiation} (1) • (bag / water) {radiates / emits} {thermal energy / heat / radiation} (1) • more heat radiated at higher temperature (1) • input and output are balanced (at steady temperature) (1)	 idea of energy input e.g. "sun heats the bag up" idea of energy output idea of more heat lost (to surroundings)at higher temperature "absorbing heat at same rate as radiating heat" (3) **ignore**(sun) light / rays	**(3)**

Q12.

Question Number	Answer	Additional guidance	Mark
(i)	any value between 21 and 54 inclusive		(1) AO3

Question Number	Answer	Additional guidance	Mark
(ii)	any one from distance / separation (1) time (exposure) (1)		(1) AO3

Question Number	Answer	Additional guidance	Mark
(iii)	Any one from poor conductor / reduces energy loss / (good) insulator (1) same height as detector (1)	allow heat loss for energy loss accept wood stops heat loss accept it does not conduct heat	(1) AO3

Q13.

	Answer	Acceptable answers	Mark
(i)	(black is the) best absorber (of radiation/heat)(compared to other colours) (1)	(Black/it) absorbs more (Black/it) is a good absorber (of radiation/heat) (black is) good at taking in (radiation/heat) Ignore (black is a) good emitter	(1)

		Ignore light. Reject black attracts heat/radiation	
(ii)	• Heating effect/temperature greatest at/beyond red (1) • (There is) radiation beyond red (end of spectrum)(1)	(idea of) different colours have different heating effects (Radiation from) sunlight causes a heating effect accept reverse argument Infrared/IR (beyond red end of spectrum)	**(2)**
(iii)	Any **one**from • To check the thermometers produce the same readings/ temperature (under the same conditions) (1) • To show that temperature changes.	(To check) they were all at the same temperature (before starting the experiment.) To be able to make a comparison (between shade and light) (To allow them to carry out a) fair test	**(1)**